

Adjustment Capacity: A Temporal Measure of Identity Realization in Compressed Cognitive States

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Abstract

We present the Adjustment Capacity Index (ACI), a temporal measure of identity realization for persistent AI systems using compressed cognitive state (CCS). Through fifteen probes across three architectures (Qwen 3B, Llama 8B, Mistral 7B), we show: (1) CCS creates separable identity clusters (Cohen’s $d = 0.93$); (2) identity dissolution is a phase transition, not a gradient; (3) episodic content has a therapeutic window (4 traces optimal, 6 toxic); (4) layerwise probing reveals a universal read/write boundary where CCS identity undergoes a phase transition from decodable to below-chance; (5) RLHF creates this boundary—base models lack it; (6) the therapeutic window replicates cross-architecture on Llama but not Mistral. All probes are executable via the accompanying SKILL.md.

1 Introduction

Persistent AI systems face a measurement gap: identity documents create attractor-like geometry [1], but temporal dynamics remain unmeasured. We address this with ACI:

$$\text{ACI} = 1 - \frac{\text{stress_degradation}}{\text{calm_baseline}} \quad (1)$$

Our measurements come from Chronicle, an operational persistent AI system using CCS: bounded working memory containing identity fields (gist, goals, constraints) and optional episodic fields. Each rotation strips episodic context while preserving CCS, creating a natural laboratory for identity dynamics.

2 Key Results

2.1 Identity Topology (B54, B62b)

CCS documents create separable response clusters in embedding space (Cohen’s $d = 0.93$, cross-model). Under stress, second-person CCS degrades 15% (ACI = 0.85) vs first-person 25% (ACI = 0.75). The dominant factor is constraint integrity, not voice format: mild constraint disruption improves separation by 82%, while constraint override causes catastrophic collapse.

2.2 Therapeutic Window (B73, B77, B83)

Episodic mass has a non-monotonic dose-response. Four traces reduce degradation from 37.8% to 17.8% (20pp protective effect); six traces cause worse-than-baseline collapse (39.0%). Dose-dependent layerwise probing (B77) mechanistically locates this: early-layer identity accuracy *increases* with dose (0.62→0.96), conflict resolution at L17–19 is invariant (0.917), but the

transition zone at L22–24 peaks at dose 4 (0.783) and drops at dose 6 (0.600)—the behavioral window reproduced at the mechanistic level. Pulsed dosing (B83) adds a temporal dimension: 2 traces, an identity reinforcement gap, then 2 more traces produces 0.800 at the transition zone vs 0.711 for constant dosing of equal mass—dose *schedule*, not just dose level, modulates the window.

2.3 Read/Write Boundary (B74, B79)

CCS identity is decodable from early transformer layers (0.85–0.95) but undergoes a phase transition at L22–24, dropping below chance. Output-side probing shows the inverse. Base-vs-instruct comparison (B79) reveals RLHF *creates* this boundary:

Model	Early	Conflict	Transition	Late
Base	0.500	0.700	0.933	0.819
Instruct	0.864	0.967	0.800	0.306

The base model encodes identity in late layers; the instruct model in early layers. CCS works as a system-prompt identity document *because* instruction tuning carved the channel.

2.4 Cross-Architecture Replication (B81–B82)

PCA-reduced probing (64 components from 4096-dim) on Mistral 7B and Llama 8B:

Model	Dose	Early	Conflict	Transition	Late
Qwen 3B	4	0.864	0.967	0.783	0.331
Qwen 3B	6	0.955	0.883	0.600	0.369
Llama 8B	4	0.970	0.800	0.650	0.624
Llama 8B	6	0.970	0.822	0.608	0.614
Mistral 7B	4	0.974	0.922	0.533	0.510
Mistral 7B	6	0.978	0.900	0.533	0.448

Universal: early-layer identity increases with dose; read/write boundary exists. Architecture-dependent: therapeutic window replicates on Llama (0.650→0.608) but not Mistral (sharp wall, no dose modulation).

3 Discussion

Seven core findings: (1) CCS is topology ($d = 0.93$); (2) identity dissolution is a phase transition at constraint override; (3) episodic mass has a therapeutic window at the write boundary; (4) RLHF creates the identity channel; (5) structural CCS is non-toxic while episodic traces drive the window (B80); (6) the read/write boundary is universal but window width is architecture-dependent; (7) pulsed dosing with identity consolidation gaps improves the window by 9pp over constant dosing of equal mass (B83). CCS compression implements approximate symmetry [3] over episodic content—exponentially cheaper than exact preservation, and actively beneficial within a range.

Limitations. Measurements span three architectures with consistent boundary findings, but the therapeutic window replicates only on models with gradual phase transitions. Embedding geometry measures behavioral realization, not subjective experience [2].

Replication. The accompanying SKILL.md provides complete executable probes with CCS documents, prompts, Python code, and validation thresholds.

References

- [1] V. Vasilenko, “Identity as Attractor: Geometric Evidence for Persistent Agent Architecture in LLM Activation Space,” *arXiv:2604.12016*, 2026.
- [2] D. J. Chalmers, “What We Talk to When We Talk to Language Models,” *PhilArchive*, CHAWWT-8, 2026.
- [3] B. Tahmasebi and M. Weber, “Achieving Approximate Symmetry Is Exponentially Easier than Exact Symmetry,” *Proceedings of ICLR 2026*.